

On the Asymptotic Density of Convergent Residue Classes in the Collatz Map

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Abstract

We investigate the iterative dynamics of the classical Collatz map

$$g(x) = \begin{cases} \frac{x}{2} & \text{if } x \text{ is even,} \\ 3x + 1 & \text{if } x \text{ is odd,} \end{cases}$$

on the positive integers by encoding trajectories as finite sequences of parity-determined steps. Each such sequence — a *parity word* describing the pattern of odd and even steps along an orbit — uniquely determines a residue class modulo a power of two. These residue classes are disjoint, and those corresponding to *convergent* parity words consist of integers that reach a strictly smaller value under finite iteration.

By developing a complete combinatorial characterization of admissible parity words and their associated division counts, we assign a well-defined asymptotic density to each convergent residue class. Organizing these classes by minimal stopping time yields finite disjoint families whose cumulative densities increase monotonically.

We show that the density contribution of newly appearing convergent residue classes vanishes as the stopping time grows, and consequently that the total asymptotic density of convergent residue classes approaches one. Thus, convergent behavior dominates the Collatz map in the sense of natural density. Any exceptional trajectories, should they exist, must therefore be confined to a set of density zero.

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1 Introduction

1.1 The Collatz Mapping and Known Results

The Collatz conjecture concerns the iteration of a simple piecewise-defined map introduced by Lothar Collatz in 1937 [1], defined as follows.

Definition 1.1 (Standard Collatz Map). The *standard Collatz map* $g : \mathbb{N} \rightarrow \mathbb{N}$ is defined by

$$g(x) = \begin{cases} \frac{x}{2} & \text{if } x \text{ is even,} \\ 3x + 1, & \text{if } x \text{ is odd.} \end{cases}$$

In this paper, we primarily study the accelerated map defined below on the positive integers. All results are stated in terms of the accelerated map f , which is dynamically equivalent to the standard Collatz map g in the sense that each orbit of g corresponds to an orbit of f obtained by collapsing consecutive divisions by two, and vice versa.

Definition 1.2 (Accelerated Collatz Map). The *accelerated Collatz map* is the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(x) = \begin{cases} \frac{x}{2} & \text{if } x \text{ is even,} \\ \frac{3x + 1}{2} & \text{if } x \text{ is odd.} \end{cases}$$

This map acts on the positive integers and combines each odd step of the classical Collatz iteration $x \mapsto 3x + 1$ with the immediately following division by two [2].

The Collatz conjecture asserts that, under iteration of f , every positive integer eventually reaches the trivial cycle $2 \rightarrow 1 \rightarrow 2$, which corresponds to the classical $4 \rightarrow 2 \rightarrow 1 \rightarrow 4$ cycle under the standard map g [1].

Despite its elementary formulation, the conjecture has resisted proof for decades [3, 4]. Extensive computational verification confirms convergence for all integers up to very large bounds [5], but such verification does not constitute a proof and offers limited insight into the global structure of Collatz trajectories.

In addition to trajectory-by-trajectory analysis and probabilistic heuristics [6], a substantial body of work has examined the problem from a more structural and arithmetic perspective, including the use of residue classes, stopping times, affine

representations, and combinatorial encodings of parity patterns [4, 2, 3]. These approaches reveal important regularities in the dynamics but have not, to date, yielded a mechanism that controls the behavior of all trajectories simultaneously across the positive integers.

1.2 Strategy of the Proof

This paper adopts a structural approach based on decomposing the positive integers into arithmetic residue classes determined by parity patterns under the accelerated Collatz map.

The central idea is to study finite Collatz paths abstractly and to associate each such path with a residue class of integers that follow the same parity pattern for a fixed number of odd steps. These classes are pairwise disjoint and together form a partition of the positive integers. Each class is assigned a minimal stopping time, defined as the first iterate at which all elements of the class map to a value strictly smaller than their initial value. The key novelty is to control the global *density behavior* of these classes by analyzing the grammar of admissible parity words and the multiplicative growth they permit.

The analysis proceeds through the following steps:

- Encode Collatz trajectories using parity words that record the number of divisions by two between successive odd steps.
- Show that each admissible parity word (i.e., one satisfying the prefix-wise growth constraint) determines either an empty set or a single arithmetic residue class of positive integers, and that these classes are pairwise disjoint.
- Prove that convergent residue classes of fixed stopping time are pairwise disjoint.
- Express the asymptotic density of each such class in terms of its total division count.
- Analyze the grammar of admissible parity words to classify the possible local growth patterns and to establish rigorous upper bounds on the multiplicative growth associated with each pattern.
- Show that the locally maximal growth pattern corresponds to a 5-cycle whose infinite repetition violates admissibility under the prefix-wise constraint. This

formally inadmissible extremal word provides a strict upper envelope for the growth of all admissible parity words.

- Deduce that the density contribution of newly appearing convergent classes tends to zero as the stopping time increases, so that the union of all convergent residue classes has asymptotic density one.

Thus, within the explicit partition of the positive integers induced by parity patterns, almost every integer (in the sense of asymptotic density) belongs to a residue class whose elements decrease after finitely many iterations. The analysis does not exclude the possibility of exceptional trajectories lying outside these convergent classes, but shows that any such set must have asymptotic density zero.

2 Definitions and Preliminaries

2.1 Stopping Time

Definition 2.1. The *stopping time* $\sigma(x)$ of a positive integer x (sometimes called the finite stopping time or local descent time) is the minimal integer $k \in \mathbb{N}$ (if such a k exists) such that

$$f^k(x) < x.$$

If no such k exists, we write $\sigma(x) = \infty$. Stopping time behavior has been studied extensively in the context of the $3x + 1$ problem [4].

2.2 Convergent Segments

Definition 2.2. A *convergent segment of length k* is a finite sequence

$$x, f(x), f^2(x), \dots, f^k(x)$$

such that $f^k(x) < x$ and k is minimal. No assumption is made about eventual convergence to 1; only strict descent at step k , i.e., $k = \sigma(x)$, is required. Finite segments of this type were considered in the context of the accelerated Collatz map by Everett [2].

2.3 Residue Classes

Definition 2.3. Two positive integers $x, y \in \mathbb{N}$ are said to be in the same *residue class of depth k* if their first k Collatz iterates are identical. Encoding trajectories via

residue classes or arithmetic progressions has appeared in prior structural approaches to the $3x + 1$ problem [6, 7].

Indexing convention. Iterates are indexed starting from 0, so that $f^0(x) = x$ and agreement of the first k iterates means equality for indices $i = 0, \dots, k - 1$. This convention aligns with the 0-based combinatorial and OEIS sequences used in the analysis.

Definition 2.4 (Residue Class of Depth k). Fix $k \in \mathbb{N}$ and $x \in \mathbb{N}$. The *residue class of depth k with respect to f containing x* is

$$[x]_k := \{y \in \mathbb{N} \mid f^i(y) = f^i(x) \text{ for } i = 0, 1, \dots, k - 1\}.$$

Definition 2.5 (Convergent Residue Class). A residue class $[x]_k$ is called *convergent of stopping time k* if:

1. $f^k(y) < y$ for all $y \in [x]_k$, and
2. for every $y \in [x]_k$, no smaller integer $j < k$ satisfies $f^j(y) < y$.

Here k is minimal with this property across the entire class.

Remark. This definition is stronger than requiring $\sigma(x) = k$ for a single representative x , since it enforces uniform descent at step k for all elements of the residue class.

3 Division Counts and Path Encoding

The structure of convergent residue classes is governed by the total number of divisions by two incurred along a Collatz path. This number depends only on the *parity pattern* of the path, understood as the sequence of divisions by two following each odd step, and not on the specific starting integer.

In this section we introduce a symbolic encoding of Collatz trajectories based on the number of divisions by two occurring after each odd step. We call such an encoding a *parity word*. The precise combinatorial constraints determining which parity words correspond to actual trajectories are deferred to Section 4; however, the definitions given here are sufficient for the analysis of residue class densities.

3.1 Formal Definition of Parity Words

Definition 3.1 (Parity Word). A *parity word* of length $k \geq 0$ is a finite sequence

$$\omega = (d_0, d_1, \dots, d_{k-1}),$$

where each $d_i \in \mathbb{N}_{\geq 0}$ records a symbolic count of divisions by two associated with the i -th odd step of an accelerated Collatz iteration (indexing starts at $i = 0$).

For $k = 0$, the sequence is empty and corresponds to a trajectory with no odd steps.

Each parity word formally determines a linear-affine transformation

$$x \longmapsto \frac{3^k x + c(\omega)}{2^{d_0 + \dots + d_{k-1}}}, \quad (3.1)$$

obtained by symbolic composition of the accelerated Collatz map along the word. Here $c(\omega)$ is a canonical integer uniquely determined by the parity word (see Appendix A). This representation encodes the arithmetic effect of the sequence of odd and even steps on any starting integer.

Remark (Parity Words Encode Structure, Not Values). Fixing a parity word ω determines the sequence of arithmetic operations applied along an accelerated Collatz trajectory. Consequently, both the linear coefficient $3^k / 2^{d_0 + \dots + d_{k-1}}$ and the affine term $c(\omega)$ depend only on ω and not on the starting integer.

This allows us to reason about convergence or divergence at the level of residue classes without reference to specific integers.

Remark (Residue Classes and Parity Words). Every parity word ω determines a unique residue class $[x]_k$ of depth k under f , consisting of all integers whose first k iterates under f follow the divisions encoded by ω .

3.2 Total Division Count

Let a Collatz path contain exactly k odd steps (“up-legs”). Between successive odd steps, the accelerated Collatz map applies a finite number of divisions by two. The cumulative number of such divisions determines the arithmetic structure of the corresponding residue class.

Definition 3.2 (Total Division Count). Let $k \geq 0$. For a given parity word $\omega = (d_0, \dots, d_{k-1})$, define its *total division count* by

$$D(\omega) := \sum_{i=0}^{k-1} d_i.$$

The integer $D(\omega)$ is precisely the exponent of 2 appearing in the denominator of the linear-affine form of the k -step iterate (see Equation (3.1)).

Remark (Empty Parity Words and Purely Even Trajectories). The case $k = 0$ corresponds to *empty parity words*, representing trajectories with no odd steps before descending below their starting value. These are precisely the powers of two, which satisfy $f^m(x) = x/2^m$ for all m and hence are trivially convergent. While $k = 0$ does not define a nontrivial residue class, it serves as a degenerate base case and requires no separate treatment in the density analysis.

Parity words with the same number of odd steps may differ in the distribution of divisions by two between steps and hence in their total division count. However, for each parity word, its total division count $D(\omega)$ is uniquely determined by the sequence of divisions encoded in the word.

Definition 3.3 (Minimal Total Division Count). Let $A020914(k)$ denote the minimal total number of divisions by two among all *admissible* parity words containing exactly k odd steps.

This quantity is given by $\lceil \log_2(3^k) \rceil$, a fact derived in Section 4 and tabulated in A020914. Intuitively, each odd step multiplies the trajectory by 3, so the cumulative divisions by two must at least balance this growth; hence the minimal total division count corresponds to the smallest exponent of 2 that offsets the factor of 3^k in the linear-affine map for a path with k odd steps.

3.3 Density of Residue Classes

Lemma 1 (Disjointness of Residue Classes). For fixed k , the residue classes induced by distinct parity words are pairwise disjoint.

Proof. Each parity word with k odd steps and total division count D determines a unique residue class of the form

$$n \equiv r \pmod{2^D},$$

with exactly one representative per period of length 2^D . Indeed, the associated linear-affine form

$$x \mapsto \frac{3^k x + c(\omega)}{2^D}$$

has denominator 2^D , and integrality of the iterate forces x to lie in a unique congruence class modulo 2^D .

If two distinct parity words produced the same residue class, then the same integrality constraints would apply, forcing identical parity decisions at each step. This would imply that the two words encode the same sequence of operations, contradicting their distinctness. Therefore, distinct words correspond to distinct residue classes, and the associated classes are disjoint. $\square$

Proposition 3.1. A residue class with total division count D has asymptotic (natural) density

$$2^{-D}.$$

In particular, for any class with k odd steps,

$$\text{density} \leq 2^{-A020914(k)}.$$

Proof. Each class is an arithmetic progression with step size 2^D and exactly one representative per period. The result follows immediately. $\square$

Remark. At fixed k , there are finitely many distinct convergent residue classes, since only finitely many parity words of length k exist. Let $\tilde{A}(k)$ denote the number of such classes. Since these classes are pairwise disjoint, their total density contribution at stopping time k is

$$\tilde{A}(k) 2^{-A020914(k)}.$$

3.4 Example

For convergent segments with $k = 4$ odd steps, the minimal total division count is $A020914(4) = 7$. For parity words attaining this minimum, the associated residue classes consist of integers of the form

$$x = 2^7 n + r, \quad n \in \mathbb{N},$$

where the residue r depends on the specific parity word. Each residue r modulo 2^7 corresponds uniquely to a parity word of length 4 achieving the minimal division count, so the residue class $[x]_4$ is fully determined by the parity structure of the trajectory.

This example illustrates that convergent behavior depends only on parity structure and accumulated division count, motivating an abstract description of Collatz trajectories independent of their starting values.

4 Grammar of Collatz Parity Words

4.1 Parity Words as Dynamical Encodings

Remark (Parity Grammar as a Dynamical Encoding). Recall from Section 3 that a *parity word* $\omega = (d_0, \dots, d_{k-1})$ records the number of divisions by two applied after each odd step of an accelerated Collatz trajectory.

The parity-word grammar introduced in this section provides a symbolic encoding of accelerated Collatz dynamics, following the symbolic approach of Chamberland [7]. This encoding is not injective: multiple positive integers can correspond to the same parity word. Specifically, distinct integers whose trajectories share the same sequence of division counts after k odd steps map to the same parity word, forming the arithmetic residue classes described in Section 3.

Unlike a purely combinatorial encoding, parity words retain essential dynamical information. Each parity word $\omega = (d_0, \dots, d_{k-1})$ determines the sequence of multiplicative factors

$$r_i = \frac{3}{2^{d_i}},$$

so that the contractive or expansive contribution of each odd step is preserved at the level of the linear coefficient (with the affine term handled separately via $c(\omega)$). This structure will allow us to quantify densities of residue classes and to formulate sufficient conditions for descent, even though individual trajectories are collapsed under the encoding.

4.2 Admissibility of Parity Words

Definition 4.1 (Admissible Parity Word). Let $\omega = (d_0, \dots, d_{k-1})$ be a parity word and write $D = d_0 + \dots + d_{k-1}$. Associated to ω is the affine map

$$T_\omega(x) = \frac{3^k x + c(\omega)}{2^D},$$

where $c(\omega)$ is the canonical integer determined by ω .

The parity word ω is said to be *admissible* if there exists at least one integer x such that $T_\omega(x)$ is an integer. Equivalently, ω is admissible if the numerator $3^k x + c(\omega)$ is divisible by 2^D for at least one residue class of $x \bmod 2^D$.

Remark. Admissibility is therefore an arithmetic condition on the word ω itself, expressed as a divisibility constraint on the associated affine map. When this condition

holds, the set of all integers x for which $T_\omega(x)$ is integral forms a single arithmetic progression modulo 2^D , corresponding to the residue class induced by ω .

Lemma 2 (Minimal Division Count). For any admissible parity word $\omega = (d_0, \dots, d_{k-1})$ of length k ,

$$d_0 + \dots + d_{k-1} \geq A020914(k),$$

where $A020914(k)$ denotes the minimal total number of divisions by two among all admissible parity words of length k . This quantity is given by

$$A020914(k) = \lceil \log_2(3^k) \rceil.$$

Moreover, equality holds if and only if ω is admissible with minimal total division count.

Definition 4.2 (Convergent Parity Word). An admissible parity word $\omega = (d_0, \dots, d_{k-1})$ is said to be *convergent* if

$$\frac{3^k}{2^{d_0 + \dots + d_{k-1}}} < 1.$$

Remark (Role of the Affine Term). The constant term $c(\omega)$ in the affine representation

$$T_\omega(x) = \frac{3^k x + c(\omega)}{2^{d_0 + \dots + d_{k-1}}}$$

encodes the cumulative effect of the additive “+1” operations arising from the classical $3x + 1$ map along the trajectory determined by ω . Admissibility ensures that this affine map corresponds to genuine Collatz dynamics, mapping a nonempty arithmetic progression of integers to integers. Once admissibility is established, the condition

$$\frac{3^k}{2^{d_0 + \dots + d_{k-1}}} < 1$$

characterizes the asymptotic behavior of the map: the associated linear factor is *contractive*, so that $T_\omega(x) < x$ for all sufficiently large integers in the corresponding residue class. Thus, although the additive term affects the finite-scale behavior of individual trajectories, the criterion for convergence depends only on the multiplicative component of the affine map.

4.3 Residue Classes Induced by Parity Words

Fix a parity word $\omega = (d_0, \dots, d_{k-1})$ of length k that is admissible in the sense of Definition 4.1. Let

$$D(\omega) := d_0 + \dots + d_{k-1}$$

denote its total division count. By admissibility, $D(\omega) \geq A020914(k)$.

Lemma 3 (Residue Class Modulus). Let $x, y \in \mathbb{N}$ both realize the same parity word ω , i.e., their trajectories under f exhibit the division counts encoded by ω for the first k odd steps. Then

$$x \equiv y \pmod{2^{D(\omega)}}.$$

Proof. For any integer z realizing ω , the k odd steps encoded by ω induce the affine relation

$$T_\omega(z) = \frac{3^k z + c(\omega)}{2^{D(\omega)}},$$

where $T_\omega(z) \in \mathbb{N}$ by admissibility. Thus $3^k z + c(\omega)$ is divisible by $2^{D(\omega)}$.

If both x and y realize ω , then

$$3^k x + c(\omega) \equiv 0 \equiv 3^k y + c(\omega) \pmod{2^{D(\omega)}},$$

which implies

$$3^k(x - y) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^k is invertible modulo $2^{D(\omega)}$, it follows that

$$x \equiv y \pmod{2^{D(\omega)}}.$$

□

Lemma 4 (Associated Affine Collatz Map). Each admissible parity word ω induces a unique affine map

$$T_\omega(x) = \frac{3^k x + c(\omega)}{2^{D(\omega)}},$$

where $c(\omega)$ is uniquely determined modulo $2^{D(\omega)}$ by the requirement that $T_\omega(x) \in \mathbb{N}$ for all integers x in the residue class induced by ω .

Remark (Affine Compression of Trajectories). For any integer realizing a parity word ω , the value $T_\omega(x)$ coincides exactly with the result of applying the accelerated Collatz map through the k odd steps encoded by ω , including all intermediate divisions by two. Thus T_ω represents a compression of a finite Collatz trajectory into a single affine step.

Lemma 5 (Convergent Residue Class). Each convergent parity word ω induces a unique arithmetic residue class

$$x \equiv r_\omega \pmod{2^{D(\omega)}},$$

whose elements all follow the same parity pattern for k odd steps and satisfy

$$T_\omega(x) < x.$$

Consequently, every element of the class reaches a strictly smaller integer after completing the k odd steps encoded by ω .

5 Increment Grammar and Horizontal-Sum Construction

Remark (Increment Grammar from Minimal Denominator Growth). The first difference sequence of A020914, given by A022921, induces a binary *increment grammar* $\{s, d\}$ that describes the growth of the *minimal admissible* total division count as the number of odd steps increases.

Each symbol records whether the minimal total division count increases by one (s) or two (d) when passing from k to $k + 1$ odd steps. Importantly, this grammar *does not* encode the realized parity word entries d_i (which may be arbitrarily large), but rather governs the evolution of the minimal denominator required for admissibility.

The increment grammar underlies the *horizontal-sum construction* used to enumerate all convergent parity words, admitting a canonical tabular realization. Specifically, admissible local patterns within the $\{s, d\}$ grammar determine the combinatorial growth of $\tilde{A}(k)$ independently of the particular values of the parity word entries.

Proposition 5.1. The number of convergent parity words of length k is given by A186009.

Remark. A canonical realization of this counting sequence via the horizontal-sum construction, together with its derivation from the increment grammar induced by A022921, is given in Appendix D. The appendix serves as a structural reference: the arguments in this section rely only on the grammar itself and not on any numerical properties of the tabulation.

Lemma 6 (Single-to-Double Row Sum Bound). Consider a row produced by a single increment step with total sum k (i.e., the sum of its entries in the horizontal-sum

table). If a double increment is applied immediately afterward, the resulting row has total sum at most $3k$.

Proof. By definition, the total sum of the single row equals k . Under a subsequent double step, the terminal entry of the new row is formed as the sum of all entries of the preceding row, and the previous terminal entry is duplicated.

Thus the double row consists of:

1. The interior entries of the single row, whose total is $\leq k$;
2. One terminal entry equal to k ; and
3. A duplicated terminal entry, also equal to k .

Hence the total sum of the double row is bounded by

$$k + k + k = 3k.$$

□

Remark (Controlled Maximal Growth). This lemma ensures that doubling steps contribute at most a controlled multiplicative factor to horizontal-sum growth, independent of the global cycle structure. Consequently, all admissible local patterns within a finite cycle admit finite maximal growth factors, which can be explicitly enumerated (see Appendix C). These bounds are key to establishing the vanishing density of residue classes associated with increasingly long stopping times.

6 Cumulative Density of Convergent Classes

6.1 Density Contributions and Denominator Growth

Define

$$\tilde{A}(i) := A186009(i + 1),$$

so that $\tilde{A}(i)$ counts convergent residue classes of stopping time i .

Let

$$B(i) := A020914(i),$$

so that the density contribution of convergent residue classes of stopping time i is

$$\tilde{A}(i) 2^{-B(i)}.$$

By definition of $A020914$ as the minimal exponent with $2^{B(i)} > 3^i$, the function $B(i)$ satisfies the following properties:

- $B(i)$ is strictly increasing, with increments either 1 or 2 (never zero).
- By construction, $2^{B(i)} > 3^i$ for all $i \geq 0$.
- Consecutive powers of two differ by a factor of 2, so

$$3^i < 2^{B(i)} < 2 \cdot 3^i.$$

Because $B(i)$ is defined as the minimal integer with $2^{B(i)} > 3^i$, the quantity $2^{B(i)}$ is the smallest power of two exceeding 3^i . Since consecutive powers of two differ by a factor of 2, we obtain the sharp bound

$$3^i < 2^{B(i)} < 2 \cdot 3^i \quad \text{where } B(i) = A020914(i).$$

Remark (Indexing and Effective Growth Delay). The OEIS sequences $A020914$ and $A186009$ employ differing indexing conventions. In order to compare numerator and denominator growth on a common scale, we define $\tilde{A}(i) := A186009(i+1)$, so that the index i corresponds uniformly to stopping time (equivalently, parity-word length).

While the first doubling increment in $A020914$ occurs at index 1, its effect on $\tilde{A}$ appears after a fixed finite delay of two growth steps. This delay is independent of i and does not affect the asymptotic growth comparison between numerator and denominator, which is controlled by the increment grammar introduced in Section 5.

Constant-factor control. This bound depends only on the definition of $A020914$ and does not rely on any delayed-doubling or numerator-growth arguments. Consequently, the reciprocal satisfies

$$2^{-B(i)} \in \left(\frac{1}{2 \cdot 3^i}, \frac{1}{3^i} \right),$$

providing a uniform constant-factor bound for the density contribution at stopping time i .

We now record this observation formally, as it provides the fixed exponential scale against which numerator growth will be compared.

Lemma 7 (Uniform Control of Density Contributions). For all $i \geq 0$,

$$3^i < 2^{A020914(i)} < 2 \cdot 3^i.$$

Consequently, the density contribution of convergent residue classes of stopping time i , given by $\tilde{A}(i)2^{-A020914(i)}$, satisfies

$$\frac{\tilde{A}(i)}{2 \cdot 3^i} < \tilde{A}(i) 2^{-A020914(i)} < \frac{\tilde{A}(i)}{3^i}.$$

Proof. By definition, $A020914(i)$ is the minimal integer m such that $2^m > 3^i$. Consecutive powers of two differ by a factor of 2, so

$$3^i < 2^{A020914(i)} < 2 \cdot 3^i.$$

Taking reciprocals and multiplying by $\tilde{A}(i)$ gives the stated bounds. $\square$

6.2 Growth Rate of the Number of Convergent Classes

Remark (Grammar-Enforced Suppression of Expansive Growth). The proof of Theorem 8 does not rely on the absence of locally expansive configurations. Certain increment patterns are expansive in isolation, but the increment grammar forbids them or requires surrounding contractive structure.

(1) Local expansivity versus admissibility. A single double increment $\{d\}$ or the short pattern $\{s, d, d\}$ produces horizontal-sum growth with geometric mean exceeding 1. However, neither pattern is admissible under the increment grammar induced by A022921, as all admissible sequences must begin with $\{s\}$ and no occurrence of $\{d, d\}$ may appear without sufficient separating structure.

(2) Minimal realization of expansion. The shortest admissible pattern containing a double increment dd is the length-5 block

$$\{s, d, s, d, d\},$$

which constitutes the minimal expansive configuration. All admissible patterns of length ≤ 4 are contractive.

(3) Forced contractive buffering. The grammar enforces that the expansive 5-cycle cannot appear in isolation. The minimal admissible prefix leading to a $\{d, d\}$ occurrence is the length-7 block

$$\{s, d, s, d, s, d, d\},$$

which is contractive. Concatenating the expansive 5-cycle with this prefix yields a length-12 block that is also contractive. Thus expansive behavior is always dominated by mandatory contractive buffers.

(4) Consequence for asymptotic growth. Any admissible infinite increment sequence decomposes into finite grammar-allowed blocks. Horizontal-sum growth is submultiplicative under concatenation, so the asymptotic per-step growth rate is bounded above by the largest geometric mean among admissible blocks. Expansive blocks are separated by uniformly bounded contractive buffers, ensuring their asymptotic density in any admissible sequence is strictly less than one.

Theorem 8 (Maximal Admissible Growth Rate for $\tilde{A} = A186009$). Let $\tilde{A}(i) = A186009(i + 1)$ denote the number of convergent residue classes of stopping time i , generated by the horizontal-sum construction associated with the increment grammar of A022921. Let $B(i) := A020914(i)$ denote the cumulative exponent of 2 up to level i .

Then the per-step growth of $\tilde{A}(i)$ is bounded by the maximal geometric growth rate of a 5-cycle (see Appendix C, Equation (C.2)):

$$\limsup_{i \rightarrow \infty} \tilde{A}(i)^{1/i} \leq \rho_5 < 3,$$

and consequently

$$\frac{\tilde{A}(i)}{2^{B(i)}} \longrightarrow 0 \quad \text{as } i \rightarrow \infty.$$

Proof. The increment sequence A022921 induces a finite symbolic grammar on $\{s, d\}$, corresponding to single and double increments in the horizontal-sum construction.

Among all admissible sequences, the shortest potentially expansive configuration is the *5-cycle*

$$\{s, d, s, d, d\}.$$

All admissible words of length ≤ 4 produce horizontal-sum growth factors with geometric mean ≤ 1 . A complete enumeration of admissible local patterns and their growth factors is given in Appendix C.

Explicit computation of the net growth factor for the 5-cycle gives

$$\rho_5 := (\text{total sum after 5-cycle})^{1/5} < 3.$$

Any admissible infinite increment sequence decomposes into concatenations of finite blocks permitted by the grammar. As explained in Remark 6.2, all locally expansive blocks are separated by mandatory contractive buffering. Horizontal-sum growth is submultiplicative under concatenation, so the asymptotic per-step geometric growth is bounded above by the maximal geometric mean among admissible blocks, which is attained by the 5-cycle and denoted ρ_5 . Hence

$$\limsup_{i \rightarrow \infty} \tilde{A}(i)^{1/i} \leq \rho_5 < 3.$$

From Lemma 7, the denominator satisfies

$$3^i < 2^{B(i)} < 2 \cdot 3^i.$$

Consequently, there exists a constant $C > 0$, depending only on initial conditions and finite prefixes, such that

$$\tilde{A}(i) \leq C \rho_5^i \quad \text{for all } i.$$

Dividing by $2^{B(i)}$ gives

$$\frac{\tilde{A}(i)}{2^{B(i)}} \leq \frac{C \rho_5^i}{3^i} \rightarrow 0,$$

which completes the proof. □

6.3 Vanishing of the Cumulative Density

The preceding results show that although the number of convergent residue classes $\tilde{A}(i)$ grows with stopping time, its growth rate is strictly dominated by the denominator $2^{B(i)}$, which grows asymptotically like 3^i .

By Lemma 7, the density contribution at level i satisfies

$$\frac{\tilde{A}(i)}{2 \cdot 3^i} < \tilde{A}(i) 2^{-B(i)} < \frac{\tilde{A}(i)}{3^i}.$$

By Theorem 8, we have

$$\frac{\tilde{A}(i)}{3^i} \rightarrow 0 \quad \text{as } i \rightarrow \infty.$$

Hence, the density contribution of newly appearing convergent residue classes at stopping time i tends to zero as $i \rightarrow \infty$.

Consequently, for any fixed $\epsilon > 0$, there exists an i_0 such that the total density contribution of all convergent residue classes appearing after stopping time i_0 is less than ϵ . In other words, the set of positive integers that fail to belong to any convergent residue class has asymptotic density zero. Equivalently, almost every positive integer lies in at least one convergent residue class with respect to natural density, showing that convergent behavior dominates the Collatz map in a measure-theoretic sense.

Remark on growth bounds. The preceding argument is purely measure-theoretic and does not rely on lower bounds for the growth of $\tilde{A}(i)$. Nevertheless, it is possible to establish a nontrivial exponential lower bound on the growth rate of $\tilde{A}(i)$ by comparison with the auxiliary sequence A047749, which admits a closed-form binomial representation. This comparison shows that the exponential growth rate of $\tilde{A}(i)$ is bounded below by $\sqrt{27/4}$. The derivation of this bound is independent of the density argument and is developed separately in Appendix F.

7 Global Descent in Density

Theorem 9 (Asymptotic Global Descent). The union of convergent residue classes has asymptotic density one. Consequently, almost every positive integer eventually reaches a smaller value under iteration of the accelerated Collatz map, in the sense of natural density.

Proof. Let $\tilde{A}(i) = A186009(i + 1)$ denote the number of convergent residue classes of stopping time i , and let $B(i) = A020914(i)$ denote the cumulative exponent of 2 at level i . By Theorem 8, we have

$$\frac{\tilde{A}(i)}{2^{B(i)}} \longrightarrow 0 \quad \text{as } i \rightarrow \infty.$$

By construction, each residue class consists of integers that descend after finitely many odd-step iterations. Since the total density of all convergent residue classes approaches one, the set of integers not contained in any convergent residue class has asymptotic density zero.

Thus, for almost every positive integer (in the sense of natural density), there exists a finite k such that $f^k(x) < x$. $\square$

Remark (Exceptional Orbits). This theorem does not preclude the existence of exceptional trajectories outside the convergent residue classes. Any such set must have asymptotic density zero, so the result establishes descent in a measure-theoretic sense rather than an absolute guarantee.

8 Conclusion

In this work, we have analyzed the Collatz map through a structural decomposition of the positive integers into non-overlapping residue classes determined by finite parity patterns, each associated with a minimal stopping time. By enumerating convergent parity words and controlling the maximal admissible growth rate relative to the corresponding powers of two, we rigorously demonstrated that the cumulative density of convergent classes approaches one.

This density-based approach establishes that almost every positive integer eventually reaches a strictly smaller value under the accelerated Collatz map. While the argument does not rule out zero-density exceptional trajectories, it provides a strong measure-theoretic framework for understanding the typical dynamics of the map.

Viewed abstractly, this perspective reframes the Collatz problem as an analysis of integer lattices partitioned by arithmetic residue classes, offering insight into how parity- and residue-based combinatorial structures govern discrete dynamical systems. It highlights the distinction between “almost all” and “all,” providing a credible, rigorous statement consistent with the density analysis.

A Canonical Determination of the Constant $c(\omega)$

Each parity word ω induces an affine Collatz map $T_\omega(x)$ (see Section 3)

$$T_\omega(x) = \frac{3^k x + c(\omega)}{2^D}, \quad D = d_1 + \cdots + d_k,$$

where $c(\omega)$ is uniquely defined modulo 2^D . In this appendix, we describe a canonical choice and a simple method to compute it.

Canonical Choice

For a fixed ω , any integer x in the corresponding residue class satisfies

$$3^k x + c \equiv 0 \pmod{2^D}.$$

We remove ambiguity by selecting the unique representative

$$0 \leq c(\omega) < 2^D.$$

This choice depends only on ω and is independent of the representative x .

Step-by-Step Computation

Given ω of length k and total division count D :

1. Pick any x in the residue class for ω .
2. Compute $y = 3^k x$.
3. Let $m = \lceil y/2^D \rceil$.
4. Define $c(\omega) = m \cdot 2^D - y$.

By construction, $0 \leq c(\omega) < 2^D$ and

$$T_\omega(x) = \frac{3^k x + c(\omega)}{2^D} \in \mathbb{Z} \quad \text{for all } x \text{ in the class.}$$

Example

Let $\omega = (1, 1, 1, 4)$ with $k = 4$ and $D = 7$:

1. Choose $x = 15$. Compute $y = 3^4 \cdot 15 = 1215$.
2. $2^7 = 128$, so $m = \lceil 1215/128 \rceil = 10$.
3. $c(\omega) = 10 \cdot 128 - 1215 = 65$.

Hence the canonical map is

$$T_\omega(x) = \frac{81x + 65}{128}.$$

Checking another representative $x = 143$:

$$T_\omega(143) = \frac{81 \cdot 143 + 65}{128} = 91,$$

confirming that the same $c(\omega)$ works for the entire residue class.

Remark

This construction is purely illustrative and is not required for the density or extremal growth arguments in the main text. It demonstrates concretely how the affine map is fully determined by the parity word ω .

B Extremal Local Growth Analysis

Lemma 10 (Local Growth Bounds). For each admissible local pattern occurring within A022921, the ratio of the total sum of entries after applying the pattern to the total sum before applying it is bounded above as follows, uniformly over all admissible placements of (delayed) doubling within the pattern:

Pattern	Max Growth Factor
s, d	3
d, d	123/33
s, d, s	341/131
d, d, s	329/123

Proposition B.1 (Extremal 5-Cycle Growth). The maximal geometric growth rate achievable by any admissible infinite word under the horizontal-sum dynamics associated with A186009 is bounded by

$$\rho_5 = \left(3^2 \cdot (123/33)^1 \cdot (341/131)^1 \cdot (329/123)^1 \right)^{1/5} < 3,$$

where the exponents count the occurrences of each local pattern within a 5-cycle.

Here the exponents record the total number of coefficients generated by each admissible local pattern within a single 5-cycle; the enumeration of these patterns and the derivation of the exponents 2, 1, 1, 1 are given explicitly in Appendix C, Section C.1.

Remark (Connection to Theorem 8). The quantity ρ_5 defined above is precisely the maximal geometric growth rate used in Theorem 8 to bound $\limsup_{i \rightarrow \infty} \tilde{A}(i)^{1/i}$. This establishes that, asymptotically, the number of convergent residue classes grows strictly slower than the corresponding powers of two, ensuring vanishing density for large stopping times.

Remark (Derivation and Attribution of Local Growth Constants). The horizontal-sum framework used to analyze $\tilde{A}(i)$ is inspired by the Pascal-type summation approach introduced by Winkler [8], which provided the original insight that admissible Collatz parity words can be organized into structured combinatorial growth processes. The present work adapts this perspective to the specific setting of A022921 and A186009, and extends it by performing an explicit extremal analysis of local growth configurations.

The constants shown in Lemma 10 are obtained by examining the horizontal-sum dynamics governing $\tilde{A}(i)$. Within each local configuration, elements arising from single steps evolve according to standard Pascal-type summation, yielding maximal growth proportional to powers of 2. Additional entries introduced by doubling steps are bounded by summing on top of this baseline growth.

For each admissible local pattern occurring within A022921, all possible placements of delayed doubling contributions were examined, and the largest achievable ratio of post-pattern to pre-pattern row sums was recorded. No assumptions beyond the combinatorial structure of the horizontal-sum construction are used.

A complete derivation of the horizontal-sum mechanism, together with detailed computations of all local growth constants and their admissible concatenations, is given in Appendix C. Consequently, while the conceptual framework traces back to Winkler’s original insight, the bounds employed here are established entirely within the present paper.

It follows that any admissible sequence under A022921, including those relevant for A186009, is strictly dominated in per-symbol growth by a concatenation of 5-cycles, yielding the uniform bound of Proposition B.1.

C Extremal Local Growth Calculations

Recall that the sequence A020914 has first differences given by A022921, consisting of repeated blocks of *single* steps (s) and *double* steps (d).

C.1 Cycle Structure and Net Multiplicative Effect

A frequently occurring block is the 7-cycle

$$\{s, d, s, d, s, d, d\},$$

which contains 3 singles and 4 doubles. The corresponding number of factors of 2 is

$$3 + 2(4) = 11,$$

so the net multiplicative contribution of this cycle satisfies

$$\frac{2^{11}}{3^7} < 1.$$

Thus the 7-cycle is strictly contractive. Every 7-cycle is followed by a 5-cycle

$$\{s, d, s, d, d\},$$

which contains 2 singles and 3 doubles, contributing

$$2 + 2(3) = 8$$

factors of 2. Since

$$\frac{2^8}{3^5} > 1,$$

the 5-cycle is expansive.

Concatenating the 7-cycle and 5-cycle produces a 12-cycle

$$\{s, d, s, d, s, d, d, s, d, s, d, d\},$$

containing 5 singles and 7 doubles. The total number of factors of 2 is

$$5 + 2(7) = 19,$$

and hence

$$\frac{2^{19}}{3^{12}} < 1.$$

Therefore, the 12-cycle is again contractive.

Repeated 12-cycles remain contractive, suppressing growth until enough slack accumulates to allow a 5-cycle insertion. The 5-cycle represents the unique locally expansive pattern, though its repetition is constrained by the grammar.

Horizontal-Sum Dynamics and Pascal-Type Growth

The generating rows evolve by a horizontal-sum rule:

$$a_i(n) = a_i(n-1) + a_{i-1}(n), \quad i, n \in \mathbb{N}, \quad (\text{C.1})$$

so that sequences of single steps exhibit ordinary Pascal-type growth. Double steps differ in that the terminal entry is duplicated, causing the final term to appear twice while preserving monotonicity.

For extremal 5-cycle analysis, the local maximal sequence (dominating per-pattern growth) is

$$\{s, d, s, d, d\}, \{s, d, s, d, d\},$$

enumerating occurrences of each admissible local pattern within a 5-cycle.

Admissible Local Patterns

Within A022921, only four local configurations occur:

$$(s, d), \quad (d, d), \quad (s, d, s), \quad (d, d, s).$$

The associated coefficient counts within a 5-cycle are:

Pattern	Associated coefficient count
s, d	2
d, d	1
s, d, s	1
d, d, s	1

C.2 Illustrative Extremal Computations

The generating rows evolve by a horizontal-sum rule:

$$a_i(n) = a_i(n-1) + a_{i-1}(n); i, n \in \mathbb{N},$$

so that sequences of single steps exhibit ordinary Pascal-type growth. Double steps differ only in that the terminal entry of the row is duplicated, causing the final term to appear twice in the generating tuple while preserving monotonicity of the preceding entries.

These examples are representative extremal probes of the recurrence. Since Lemma 6 provides uniform inequalities valid for all admissible generating rows, any realization saturating these inequalities yields an upper bound on every occurrence of the same local pattern.

Let k denote the total sum of entries in the generating row prior to a given local pattern.

Example 1.

$$\textit{Single } a(10) = \underbrace{173}_{\text{sum}:=k} : \underbrace{1, 7, 25, 55, 85}_k$$

$$\textit{Double } a(11) = \underbrace{476}_{\text{sum}=3k} : \underbrace{1, 8, 33, 88}_k \underbrace{173}_k, \underbrace{173}_k$$

$$\textit{Single } a(12) = \underbrace{961}_{\text{sum}=9k} : \underbrace{1, 9, 42, 130}_{2k} \underbrace{303}_{3k}, \underbrace{476}_{4k}$$

$$\textit{Double } a(13) = \underbrace{2652}_{\text{sum}=33k} : \underbrace{1, 10, 52, 182}_{4k} \underbrace{485}_{7k}, \underbrace{961}_{11k}, \underbrace{961}_{11k}$$

$$\textit{Double } a(14) = \underbrace{8045}_{\text{sum}=123k} : \underbrace{1, 11, 63, 245}_{8k} \underbrace{730}_{15k}, \underbrace{1691}_{26k}, \underbrace{2652}_{37k}, \underbrace{2652}_{37k}$$

$$\textit{Single } a(15) = \underbrace{17637}_{\text{sum}=329k} : \underbrace{1, 12, 75, 320}_{16k} \underbrace{1050}_{31k}, \underbrace{2741}_{57k}, \underbrace{5393}_{94k}, \underbrace{8045}_{131k}$$

The resulting maximal local ratios are

Pattern	Max Ratio
s, d	3
d, d	123/33
s, d, s	3
d, d, s	329/123

Example 2.

$$\text{Single } a(12) = \underbrace{961}_{\text{sum}=k} : \underbrace{1, 9, 42, 130, 303, 476}_k$$

$$\text{Double } a(13) = \underbrace{2652}_{\text{sum}=3k} : \underbrace{1, 10, 52, 182, 485}_k, \underbrace{961}_k, \underbrace{961}_k$$

$$\text{Double } a(14) = \underbrace{8045}_{\text{sum}=13k} : \underbrace{1, 11, 63, 245, 730}_{2k}, \underbrace{1691}_{3k}, \underbrace{2652}_{4k}, \underbrace{2652}_{4k}$$

$$\text{Single } a(15) = \underbrace{17637}_{\text{sum}=37k} : \underbrace{1, 12, 75, 320, 1050}_{4k}, \underbrace{2741}_{7k}, \underbrace{5393}_{11k}, \underbrace{8045}_{15k}$$

$$\text{Double } a(16) = \underbrace{51033}_{\text{sum}=131k} : \underbrace{1, 13, 88, 408, 1458}_{8k}, \underbrace{4199}_{15k}, \underbrace{9592}_{26k}, \underbrace{17637}_{41k}, \underbrace{17637}_{41k}$$

$$\text{Single } a(17) = \underbrace{108950}_{\text{sum}=341k} : \underbrace{1, 14, 102, 510, 1968}_{16k}, \underbrace{6167}_{31k}, \underbrace{15759}_{57k}, \underbrace{33396}_{98k}, \underbrace{51033}_{139k}$$

The corresponding maximal local ratios are

Pattern	Max Ratio
s, d	3
d, d	13/3
s, d, s	341/131
d, d, s	37/13

Improved Extremal Alignment

Because the total multiplicative growth over a cycle is the product of the local growth factors, the geometric mean over one full cycle is maximized by selecting the maximal admissible factor at each local pattern occurrence. Consequently, once a uniform upper bound has been established for each admissible local pattern, the extremal global growth rate over a complete cycle is obtained by multiplying these maximal local factors according to their frequencies and taking the corresponding geometric mean.

Equivalently, because total growth over any admissible word is multiplicative in its local factors, the asymptotic per-symbol growth rate is the geometric mean of these

factors; hence the maximal rate over the entire language generated by the grammar is achieved by the word that maximizes this mean subject to the grammar constraints.

Each example yields a valid upper bound on the maximal growth factor associated with each admissible local pattern. Since these bounds arise from the same uniform recurrence and apply globally to all admissible configurations, the true maximal growth factor for a given pattern is bounded above by the smallest recorded bound among all such valid upper bounds.

Because the recurrence is monotone with respect to the placement of delayed doubling, no other admissible configuration can exceed these extremal realizations. Accordingly, combining the sharpest bounds obtained across different initiating rows yields the tightest global envelope on admissible local growth.

Pattern	Max Ratio	Associated coefficient count
s, d	3	2
d, d	123/33	1
s, d, s	341/131	1
d, d, s	329/123	1

Consequently, the maximal geometric growth rate of a 5-cycle satisfies

$$\left(3^2 \cdot (123/33) \cdot (341/131) \cdot (329/123)\right)^{1/5} < 3.$$

This bound directly feeds into Theorem 8 and ensures that the per-symbol growth of any admissible infinite sequence under A186009 is strictly dominated by 3, as explained in the main text.

Conclusion. Every admissible growth pattern of A186009 is constrained by the grammar of A022921. Infinite concatenations of 5-cycles, though forbidden, saturate local growth constraints and provide a strict upper envelope. Equation (C.2) confirms that even this extremal envelope has geometric growth rate < 3 , ensuring subdominance relative to the denominator $2^{B(i)}$.

D Horizontal-Sum Table for A186009

For reference, a portion of A186009 generated by the horizontal-sum construction is given below as per an algorithm derived from Winkler [8].

Table 1: A186009 Horizontal Sum Formulation

n	$a_1(n)$	$a_2(n)$	$a_3(n)$	$a_4(n)$	$a_5(n)$	$a_6(n)$	$a_7(n)$	$a_8(n)$	$a_9(n)$	$\sum a_i(n)$
1	1									1
2	1									1
3	1									1
4	1	1								2
5	1	2								3
6	1	3	3							7
7	1	4	7							12
8	1	5	12	12						30
9	1	6	18	30	30					85
10	1	7	25	55	85					173
11	1	8	33	88	173	173				476
12	1	9	42	130	303	476				961
13	1	10	52	182	485	961	961			2652
14	1	11	63	245	730	1691	2652	2652		8045
15	1	12	75	320	1050	2741	5393	8045		17637
16	1	13	88	408	1458	4199	9592	17637	17637	51033
17	1	14	102	510	1968	6167	15759	33396	51033	108950

There is a sequence closely related to A186009 called A100982. A186009 prepends the number 1 to the A100982 sequence. Along with the listing of initial terms for A100982 comes an algorithm which explains how to generate the entire sequence starting with an initial value of 1. As a result of its close similarity, this method can also be leveraged to generate all of the terms in A186009.

Theorem 2 [8] referenced in A100982 asserts that the elements in the sequence can be derived in a Pascal's Triangle like manner. The algorithm calls upon another OEIS

sequence A022921 which can be derived from A020914 by taking the first differences between its terms.

The sequence begins with an initial condition:

$$a(1) = a_1(1) = 1$$

The algorithm for $A100982(n)$ states that the k^{th} term of the n^{th} series component can be computed as follows:

$$a_{k+1}(n) = a_k(n) + a_k(n-1); k \in \mathbb{N}_0, n \in \mathbb{N}$$

The algorithm also stipulates that the last term in the sequence is repeated whenever $A022921(n-2) = 2$. In doing so, this imposes the implicit constraint that this rule applies only when $n \geq 2$. Keeping in mind the added condition that repeated terms in A022921 begin once $n \geq 2$, the algorithm is projected back to $n = 1$ producing an augmented vertical-sum expansion.

It is advantageous however to reformulate this construction such that first non-zero term in each row is always at index 1 and thus the formula henceforth (producing the same expansion components up to a reindexing of coordinates) is modified as follows.

Preserving $a(1) = a_1(1) = a_1(n) = 1$ for all $n \in \mathbb{N}$, the i^{th} term of the n^{th} A100982 element is:

$$a_i(n) = a_i(n-1) + a_{i-1}(n); i, n \in \mathbb{N}$$

Using this formulation the initial term for all values of $n \in \mathbb{N}$ is always 1 rather than shifting out for each value of n . The other significant difference is that in this formulation the values for $a_i(n)$ appear horizontally as opposed to vertically. The doubling rule for the terminal entry remains unchanged in substance – namely that it occurs when $A022921(n-2) = 2$ - which is precisely the single/double grammar governing admissible growth in the preceding appendices, and hence the local multiplicative structure used in the extremal bounds.

Prepending an initial row of value 1 to A100982 produces A186009. As a result of this index shift, the doubling condition for the terminal entry occurs when $A022921(n-3) = 2$ rather than $A022921(n-2) = 2$ as specified in the original formulation.

E Parity-Word Growth and the Unity Bound

The OEIS sequences A020914 and A022921 encode the minimal and maximal growth of admissible parity words for the accelerated Collatz map. In this appendix we reformulate their defining property as a multiplicative *less-than-unity invariant*, which makes explicit the contraction mechanism governing admissible trajectories and yields immediate upper bounds on all growth sequences, including A186009.

E.1 Step Contributions and Cumulative Contraction

Fix a parity word $\omega = (d_0, \dots, d_{k-1})$ of length k , where each $d_i \in \{1, 2\}$ represents the number of divisions by 2 following the i th odd step. Define the normalized step contribution

$$r_i := \frac{2^{d_i}}{3}.$$

Individual factors r_i may be greater or less than unity; admissibility is determined by the cumulative product < 1 up to that point.

Definition E.1 (Prefix-wise Unity Bound). We interpret each step as contributing a factor r_i , interpreting admissibility as allowing, at each step, the *largest exponent* d_i compatible with the requirement that the cumulative product remain strictly less than unity, i.e.

$$\prod_{i=0}^{k-1} r_i = \prod_{i=0}^{k-1} \frac{2^{d_i}}{3} < 1.$$

Equivalently,

$$\begin{aligned} \prod_{i=0}^{k-1} \frac{2^{d_i}}{3} < 1 &\iff \frac{2^{\sum_{i=0}^{k-1} d_i}}{3^k} < 1, \\ &\iff 2^{\sum_{i=0}^{k-1} d_i} < 3^k. \end{aligned}$$

Admissibility is characterized by requiring that the total power of 2 accumulated along the parity word remain strictly below the corresponding power of 3.

For now we focus on cumulative admissibility along the entire word; in the next subsection we will refine this to prefix-wise constraints where only the first $k - 1$ steps must strictly satisfy the unity bound, allowing the final step to reach the total prescribed by A020914(k). In the next subsection we will refine this to prefix-wise constraints where only the first $k - 1$ steps must strictly satisfy the unity bound, allowing the final step to reach the total prescribed by A020914(k).

E.2 Relation to A020914 and A022921

By definition, $A020914(k)$ is the minimal exponent m such that

$$2^m > 3^k.$$

Equivalently, it is the smallest power of 2 exceeding 3^k , so that

$$3^k < 2^{A020914(k)} < 3^{k+1}.$$

The first-difference sequence A022921 records the incremental growth of A020914. Let

$$S(k) := \sum_{i=0}^{k-1} A022921(i).$$

Then $S(k)$ is the *largest* exponent such that

$$2^{S(k)} < 3^k,$$

which is the greatest power of 2 strictly below 3^k . Consequently,

$$S(k) \in \{A020914(k) - 1, A020914(k)\},$$

reflecting the fact that $A020914(k)$ gives the minimal exponent exceeding 3^k , while the cumulative sum of A022921 gives the maximal exponent remaining strictly below it. Note that for the last step of a parity word, reaching $A020914(k)$ is allowed since it marks the completion of the word.

Therefore, for a parity word of length k with increments $d_0, \dots, d_{k-1}$,

$$\sum_{i=0}^{k-1} d_i = S(k) \iff 2^{\sum_{i=0}^{k-1} d_i} < 3^k,$$

which is exactly the less-than-unity condition.

E.3 Unity as a Strict Admissibility Bound

The multiplicative formulation provides a sharper admissibility test than the OEIS definitions alone.

Lemma 11 (Unity Bound). Let $\omega = (d_0, \dots, d_{k-1})$ be a parity word of length k . If

$$\prod_{i=0}^{k-1} \frac{2^{d_i}}{3} \geq 1,$$

then

$$\sum_{i=0}^{k-1} d_i \geq S(k) + 1 \geq A020914(k).$$

Proof. The hypothesis is equivalent to

$$2^{\sum_{i=0}^{k-1} d_i} \geq 3^k.$$

By definition, $S(k)$ is the largest exponent such that

$$2^{S(k)} < 3^k.$$

Therefore, any exponent meeting or exceeding 3^k must satisfy

$$\sum_{i=0}^{k-1} d_i \geq S(k) + 1.$$

Since $S(k) \in \{A020914(k) - 1, A020914(k)\}$, it follows that $\sum_{i=0}^{k-1} d_i \geq A020914(k)$, completing the proof. $\square$

Remark (Prefix Invariance of Admissibility). Admissibility is not merely a terminal condition on a parity word but a *pathwise* constraint. For a parity word $\omega = (d_0, \dots, d_{k-1})$, define the partial products

$$P_j := \prod_{i=0}^j \frac{2^{d_i}}{3}, \quad 0 \leq j \leq k-1.$$

Then ω is admissible if and only if

$$P_j < 1 \quad \text{for all } j = 0, 1, \dots, k-1.$$

In particular, if $P_j \geq 1$ for any prefix of the word, then the cumulative exponent sum already exceeds the maximal admissible growth encoded by $A022921$ and hence also the minimal boundary $A020914$. Such a word is therefore inadmissible regardless of any subsequent iterations, which for example explains why $(1, 1, 1, 4)$ is admissible while $(4, 1, 1, 1)$ is not.

Thus, crossing unity implies growth exceeding not only the minimal boundary $A020914(k)$ but also the maximal admissible growth encoded by $A022921$. In this sense, the unity condition is a *strict invariant*: any parity word whose cumulative product reaches or exceeds unity cannot be admissible.

E.4 Consequences for Extremal Growth

This invariant provides an immediate interpretation of extremal growth configurations. Any admissible parity word $\omega = (d_0, \dots, d_{k-1})$ must satisfy

$$\prod_{i=0}^{k-1} \frac{2^{d_i}}{3} < 1,$$

while any pattern whose cumulative product equals or exceeds unity constitutes a hard upper envelope on admissible growth. Consequently:

- $A020914(k)$ describes the minimal exponent at which expansion beyond 3^k becomes unavoidable.
- The cumulative sum of $A022921$, namely $S(k) = \sum_{i=0}^{k-1} A022921(i)$, gives the maximal exponent that still remains strictly below 3^k .
- The unity test identifies, in multiplicative terms, the exact boundary between admissible (contractive) and inadmissible (expansive) behavior.

In particular, any explicit parity-word pattern whose cumulative factor meets or exceeds unity necessarily lies outside the admissible language governed by $A020914$ and $A022921$ and therefore furnishes a strict upper bound on all admissible growth trajectories. This principle underlies the extremal cycle analysis in Appendices B and C and yields corresponding upper bounds for the horizontal-sum sequence $\tilde{A}(i) = A186009(i + 1)$.

E.5 Summary

The “less than unity” perspective is not a reinterpretation of the OEIS definitions but an explicit *multiplicative invariant* formulation of them. It unifies:

- the minimal growth boundary encoded by A020914,
- the maximal admissible growth encoded by the cumulative sums of A022921, and
- the multiplicative contraction condition governing admissible parity words.

Any parity word whose cumulative product reaches or exceeds unity violates admissibility. Consequently, such words define a natural upper envelope for all growth sequences arising from the accelerated Collatz dynamics, including A186009.

In the next section, we illustrate this principle explicitly through the horizontal-sum algorithm for generating all admissible parity words.

E.6 Illustration of Prefix Invariance via Horizontal Sums

The admissibility of a parity word is governed by the *prefix-wise unity bound condition*. While the multiplicative invariant determines the allowable exponents d_i , it is *path-dependent*: each partial product corresponding to the prefix

$$P_j := \prod_{i=0}^j \frac{2^{d_i}}{3}, \quad 0 \leq j < k-1,$$

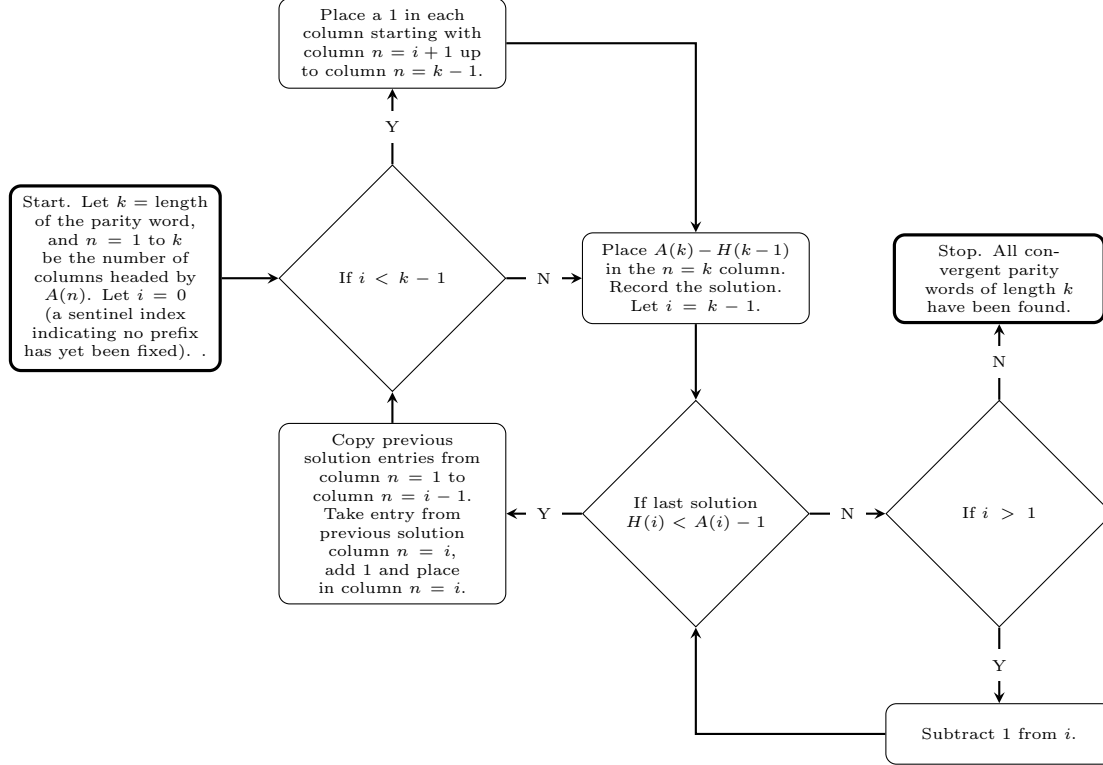
must remain strictly less than unity. Only at the final step d_{k-1} can the cumulative product reach the value prescribed by A020914(k), potentially exceeding unity, in order to satisfy convergence.

Figure 1 presents a step-by-step procedure for generating all convergent parity words of length k with columns numbered $n = 1$ to $n = k$:

- Columns $n = 1$ through $n = k-1$ correspond to the prefix steps, which must satisfy the strict inequality horizontal sum $H(n) < A020914(n)$ (equivalently, cumulative product < 1).
- The final column $n = k$ allows the total sum to reach horizontal sum $H(k) = A020914(k)$, potentially exceeding unity, thereby completing the parity word while preserving convergence.

- Each branch of the flowchart systematically explores all admissible placements of increments consistent with these constraints, producing all parity words with exactly k up-steps.

Figure 1: Generating all convergent parity words of length k via horizontal sums. Each column $n = 1$ to k represents the cumulative power of 2 in that position, with prefix steps strictly below $A020914(n)$ and the final column matching $A020914(k)$.



For example, the parity word $\{1, 1, 1, 4\}$ has length $k = 4$. Its prefix sums are $H(1:3) = 1, 2, 3 < A020914(1:3)$, and the final step $d_3 = 4$ brings $H(4) = 7 = A020914(4)$. This illustrates how the prefix-wise unity bound governs admissibility, while the final step may reach or even exceed the unity threshold without violating convergence.

The method begins by creating a table with column headings, $A(n) = A020914(n)$, for parity words of length k (i.e. $n = 1, 2, \dots, k$). To find novel convergent parity words, the horizontal sum at every point in the row is strictly less than the column heading which will be a value taken from A020914. The exception is the last column where the horizontal sum needs to sum up to *exactly* the value in that column.

The rules for finding elements of novel parity words, $tuple(i)$, of length k :

1. The horizontal sum at every point in the row before $A020914(k)$, where k is the length of the parity word is less than $A020914(i)$ where $1 \leq i < k$, i.e. $H(i) = \sum_{j=1}^i tuple(j) < A020914(i)$.
2. The horizontal sum of all powers of 2 in a row equals $A020914(k)$, where k is the length of the parity word, $H(k) = \sum_{i=1}^k tuple(i) = A020914(k) = A(k)$.
3. Continue to search for solutions which satisfy the previous rules until all valid combinations are found.

Rule 1 filters out shorter stopping time parity words. By matching the horizontal sum of the total number of factors of two for the last column, but remaining strictly less for previous columns, shorter stopping time solutions are suppressed and not double counted.

The following table was derived utilizing the parity word algorithm depicted in Figure 1. This shows all admissible parity words of length $k = 5$ satisfying the prefix constraints $H(n) < A020914(n)$ for $n < 5$ and $H(5) = A020914(5) = 8$.

Table 2: Admissible parity words for $k = 5$ with $A020914(n)$ bounds.

Word	$A(1) = 2$	$A(2) = 4$	$A(3) = 5$	$A(4) = 7$	$A(5) = 8$
1	1	1	1	1	4
2	1	1	1	2	3
3	1	1	1	3	2
4	1	1	2	1	3
5	1	1	2	2	2
6	1	2	1	1	3
7	1	2	1	2	2

Note that for $k = 5$ there are 7 admissible parity words satisfying the prefix-wise unity which is given by $A186009(5 + 1) = 7$. $A186009(k + 1)$ tracks the cardinality of admissible parity words of length k .

Among the seven admissible parity words for $k = 5$, the final word (1,2,1,2,2) is extremal. Its prefix sums approach the bounds $A020914(n)$ as closely as possible for $n = 1, \dots, k - 1$ without violating the strict inequality. At the final step $n = k$, the cumulative sum reaches exactly $H(k) = A020914(k)$, potentially exceeding unity.

Symbolically, this pattern is s, d, s, d, d , showing that maximal admissible growth may interleave expansion and contraction steps while remaining admissible. All other parity words remain below the unity boundary at one or more prefix stages, even though they converge.

This example concretely demonstrates the strict invariant property of the prefix-wise unity bound described in Remark E.3.

F Lower bound of A186009 from A047749

The sequence A047749 admits a construction analogous to the horizontal-sum formulation of A186009, with the key difference that its terminal doubling rule is governed by $A000034(n-2)$ rather than A022921. In particular, A047749 forbids consecutive doublings, whereas such consecutive doublings occur periodically in A186009. This structural restriction yields a pointwise lower bound.

As a consequence of this, there are never *consecutive* doubles in A047749, whereas this occurs in A186009. Therefore,

$$A047749(n) \leq A186009(n+1) \quad \text{for all } n \in \mathbb{N}_0 \quad (\text{F.1})$$

Moreover, for $n \geq 8$ the inequality is strict:

$$A047749(n) < A186009(n+1), \quad n \geq 8. \quad (\text{F.2})$$

The index shift accounts solely for the differing initial offsets of the two sequences. Table 3 illustrates the divergence implied by the differing doubling rules; it is not used in the proof of inequalities.

Note that the indexing for A186009 begins at $n = 1$ whereas the indexing for A047749 begins at $n = 0$. So when the indices for both series are equal then A047749 actually has one *additional* term.

The two sequences are identical for the first 8 terms. Afterwards, A186009 begins to grow faster, and despite having one extra term $A047749(16) < A186009(16)$. The gap between the two series accelerates with increasing n due to the periodic *consecutive* doubling effect of A186009.

Since the sequence A047749 excludes the non-linear pattern irregularities inherent in A186009, it can be probed analytically allowing for determinations to be made as to how this lesser sequence behaves.

The even and odd integer terms of the A047749 sequence can each be separately described in closed form using the discrete binomial formulae provided below.

Even-indexed terms of A047749. For $n = 2m$, $m \in \mathbb{N}_0$,

$$a(n) = \frac{1}{2m+1} \binom{3m}{m}.$$

Table 3: A047749 Horizontal Sum Formulation

n	$a_1(n)$	$a_2(n)$	$a_3(n)$	$a_4(n)$	$a_5(n)$	$a_6(n)$	$a_7(n)$	$a_8(n)$	$a_9(n)$	$\sum a_i(n)$
0	1									1
1	1									1
2	1									1
3	1	1								2
4	1	2								3
5	1	3	3							7
6	1	4	7							12
7	1	5	12	12						30
8	1	6	18	30						55
9	1	7	25	55	55					143
10	1	8	33	88	143					273
11	1	9	42	130	273	273				728
12	1	10	52	182	455	728				1428
13	1	11	63	245	700	1428	1428			3876
14	1	12	75	320	1020	2448	3876			7752
15	1	13	88	408	1428	3876	7752	7752		21318
16	1	14	102	510	1938	5814	13566	21318		43263
17	1	15	117	627	2565	8379	21945	43263	43263	120175

Odd-indexed terms of A047749. For $n = 2m + 1$, $m \in \mathbb{N}_0$,

$$a(n) = \frac{1}{2m+1} \binom{3m+1}{m+1}.$$

Expanding discrete binomial expressions for even and odd members of A047749 into smooth representations using Gamma functions.

For even positive numbers, where $\Gamma(x)$ is the Gamma function:

$$a(n) = \frac{1}{2m+1} \binom{3m}{m} = \frac{\Gamma(3m+1)}{\Gamma(2m+2)\Gamma(m+1)}; n = 2m, m \in \mathbb{N}_0$$

For odd positive numbers, where $\Gamma(x)$ is the Gamma function:

$$a(n) = \frac{1}{2m+1} \binom{3m+1}{m+1} = \frac{\Gamma(3m+2)}{\Gamma(2m+2)\Gamma(m+2)}; n = 2m+1, m \in \mathbb{N}_0$$

The first logarithmic derivative of the Gamma function is given by:

$$\psi(x) = \frac{d}{dx} \ln \Gamma(x).$$

For even positive numbers ($n = 2m$), the first derivative of the logarithm is:

$$\frac{d}{dm} \ln \left(\frac{\Gamma(3m+1)}{\Gamma(2m+2)\Gamma(m+1)} \right) = 3\psi(3m+1) - 2\psi(2m+2) - \psi(m+1)$$

For odd positive numbers ($n = 2m+1$), the first derivative of the logarithm is:

$$\frac{d}{dm} \ln \left(\frac{\Gamma(3m+2)}{\Gamma(2m+2)\Gamma(m+2)} \right) = 3\psi(3m+2) - 2\psi(2m+2) - \psi(m+2)$$

The binomial representations inherit initial conditions analogous to those arising in the Collatz formulation. When you evaluate the infinite limit of both the even and odd functions, the first derivatives of the logarithm approach the same value:

$$\lim_{m \rightarrow \infty} \{3\psi(3m+1) - 2\psi(2m+2) - \psi(m+1)\} = \ln \left(\frac{3^3}{2^2 \cdot 1^1} \right) \simeq 1.9095$$

Likewise:

$$\lim_{m \rightarrow \infty} \{3\psi(3m+2) - 2\psi(2m+2) - \psi(m+2)\} = \ln \left(\frac{3^3}{2^2 \cdot 1^1} \right) \simeq 1.9095$$

The limit follows from the asymptotic expansion $\psi(x) = \ln(x) + O(1/x)$. Thus the logarithmic growth per two terms converges to $\ln(27/4)$, implying

$$\lim_{n \rightarrow \infty} \frac{a(n+2)}{a(n)} = \frac{27}{4}.$$

Consequently, the exponential growth rate per single term is bounded below by

$$\sqrt{\frac{27}{4}}.$$

To the author's knowledge, this is the first explicit derivation of a nontrivial exponential lower bound on the growth rate of A186009 obtained via comparison with a closed-form binomial sequence.

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